

RAHUL AGARWAL

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TECHNICAL SKILLS

Languages: Python, Java, MySQL, HTML5, CSS3

AI, ML & Computer Vision: PyTorch, TensorFlow / Keras, YOLO (v8, v11), OpenCV, Albumentations, NumPy, Pandas, Roboflow, MakeSenseAI

LLM & GenAI: Prompt Engineering, RAG, LangChain, Grok API, AWS Bedrock

Backend & Web Frameworks: FastAPI, Flask, Streamlit, REST APIs, SQLAlchemy, Playwright, BeautifulSoup

Frontend Development: React.js, Vite, Tailwind CSS

Databases, Queues & Storage: PostgreSQL, MongoDB, MySQL, Redis, Celery

Cloud & DevOps: AWS (EC2, S3, SageMaker, Bedrock), Docker, Git, GitHub, Postman, Linux/Bash

EDUCATION

Vellore Institute Of Technology

2023 – 2027

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.82/10

EXPERIENCE

- **RS Mech Tech Pvt. Ltd.** Ahmedabad, India
Research Associate Intern May 2026 – July 2026
 - **AI & Computer Vision:** Built 4-stage ML workflows—data preprocessing, training, inference, and evaluation—using **Python, PyTorch, TensorFlow/Keras, YOLO, OpenCV, and NumPy**.
 - **Cloud Deployment:** Supported model storage and deployment using 3 AWS services—**S3, EC2, and SageMaker**—while documenting experiments and collaborating on solution improvements.
- **Mantra Softech Pvt. Ltd.** Ahmedabad, India
Computer Vision Intern May 2025 – June 2025
 - **Vision Pipelines:** Developed computer vision pipelines for 2 core tasks—object detection and recognition—using **Python, YOLO, TensorFlow, and OpenCV**.
 - **Model Optimization:** Optimized inference and deployment across 3 stages—preprocessing, detection, and post-processing—to improve real-world model performance.

PROJECTS

Lead Generation & AI Outreach Platform

[GitHub](#)

Python, FastAPI, PostgreSQL, SQLAlchemy, Playwright, BeautifulSoup, Grok API, Celery, Redis, SMTP, Streamlit

- Built a FastAPI platform to collect and manage business leads, using Playwright and BeautifulSoup to extract **7 key fields**: name, address, website, phone, rating, reviews, and description.
- Automated personalized email generation with the OpenAI API and batch delivery using Celery, Redis, and SMTP; added a Streamlit dashboard to track leads, campaign status, and delivery errors.

Real-Time Garbage Detection System

[GitHub](#)

Python, YOLOv11, OpenCV, React, Vite, Roboflow

- Trained and evaluated a YOLO-based object detector on **1000+ annotated waste images** to identify garbage classes in real time.
- Built an OpenCV inference pipeline for live video input, achieving **86.4% mAP@50** at **60 FPS** on the target device.

HeadHawk – Real-Time Head Detection System

[GitHub](#)

Python, YOLOv8, OpenCV, Streamlit, Albumentations, MakeSenseAI

- Fine-tuned a **YOLOv8** model on **700+ labeled images** for real-time head detection in video streams.
- Developed a Streamlit interface with OpenCV-based webcam/video inference, reaching **96% detection accuracy** at **24 FPS**.

CERTIFICATIONS

- [AWS Certified AI Practitioner](#) (AIF-C01) Amazon Web Services (AWS), 2026.
- [AWS Certified ML Engineer Associate](#) (MLA-C01) Amazon Web Services (AWS), 2026.